

# Jose E. Rodríguez-Ríos

Salinas, Puerto Rico | (407) 729-9500 | jerr.ccom@gmail.com  
jose.rodriguez216@upr.edu | <https://github.com/joserico00>  
<https://www.linkedin.com/in/jose-rodriguez-rios-b1b01a18a>



joserico00.github.io

*Ph.D. student and computer scientist with national-laboratory experience in cybersecurity, AI/HPC, and research software.*

## EDUCATION

|                                                                                                  |                    |
|--------------------------------------------------------------------------------------------------|--------------------|
| <b>Ph.D., Software Engineering</b> , University of Puerto Rico - Mayagüez                        | Aug 2026 - Present |
| <b>M.S., Computer Science</b> , University of Puerto Rico - Río Piedras   GPA: 4.0/4.0           | May 2025           |
| <b>B.S., Computer Science</b> , University of Puerto Rico - Río Piedras   Minor in Cybersecurity | Dec 2020           |

## TECHNICAL SKILLS

**Programming languages:** Python, C, C++, Java, Assembly  
**AI and HPC:** PyTorch, Hugging Face, TensorFlow, transformers, SLURM, GPU computing, Jupyter  
**Cybersecurity:** Vulnerability assessment, reverse engineering, Nmap, OpenVAS, Ghidra, Wireshark, Scapy  
**Systems and DevOps:** Linux, Docker, GitLab CI/CD, Git, Elasticsearch, Flask, Streamlit, SQLite, MySQL, OpenVPN

## EXPERIENCE

**Volunteer Software Developer**, Abeyno Coop solar cooperative, Salinas, PR 2024 - Present

- Built Solar Fleet Reporter, a 1,700-line Streamlit app with 88 automated checks that pulls energy data from Sol-Ark, FranklinWH, and sonnen portals, applies net-metering billing rules, and exports CSV/Excel reports and printable homeowner statements.

**Research Intern**, Argonne National Laboratory, Lemont, IL Jun 2026 - Aug 2026

- Built DNA sequence-classification workflows with PyTorch, Hugging Face, and DNABERT on Perlmutter, Polaris, and Aurora; compared 6-mer and BPE tokenization (81% vs. 80% accuracy; 195 vs. 34 tokens per sequence).
- Led a 10-student team as project lead: scoped a 10-12-hour project, wrote Jupyter/SLURM lessons and CPU-versus-GPU experiments, and coached final presentations; its fine-tuned DNABERT reached 91.7% evaluation accuracy on CTCF data.

**Graduate Research and Teaching Assistant**, UPR Río Piedras, San Juan, PR Aug 2022 - May 2025

- Developed NVSRCO, an open-source vulnerability scanner using Python, Nmap, and OpenVAS, with CSV reporting, a Dash dashboard, and WireGuard links to remote networks.
- Led Operating Systems, Computer Architecture, and Cybersecurity labs (Linux, Ghidra, Wireshark, Scapy); authored Capture the Flag (CTF) challenges in reverse engineering, cryptography, web, and forensics for about 60 Puerto Rico high-school students.

**DevOps Intern**, Lawrence Livermore National Laboratory, Livermore, CA Jun 2024 - Aug 2024

- Built a containerized Elasticsearch store for Ghidra BSim binary features so researchers can query malware-like samples at scale, with GitLab CI/CD for deployment.
- Deployed a Docker-hosted Ollama (Llama 3) service for the PaperQA PDF-reading application.

**Cybersecurity Intern**, GM Sectech, San Juan, PR Jun 2022 - Aug 2022

- Built a Python/Twilio smishing (SMS phishing) simulator for awareness training; collaborated with penetration testers on credential-theft and domain-compromise scenarios; trained on Metasploit, Hashcat, and Burp Suite.

## PROJECTS

**Malicious Traffic Detection:** Built a TensorFlow pipeline on IoT-23 logs with capture-aware splits, training-only preprocessing, and a logistic-regression baseline.

**Malware Image Classification:** Trained a TensorFlow CNN on the 9,401-image, 25-family Maling dataset to 97.7% validation accuracy.

## AWARDS AND ACTIVITIES *(4 of 7; full list in CV)*

**BSides Puerto Rico CTF (2024):** 1st place of 30 teams; reverse engineering, cryptography, exploitation.

**Tracer Fire Blue Team, ACSAC (2019):** 2nd place team of 40 participants; SIEM/IDS, Autopsy, Ghidra.

**Bloomberg HSI Tech Summit (2024):** 1 of 29 selected from 400+ applicants; Summer Accelerator invitee.

**Peer Mentor, Intro to HPC Bootcamp, Argonne (2025):** mentored 5 students designing HPC under a 12 MW cap.